

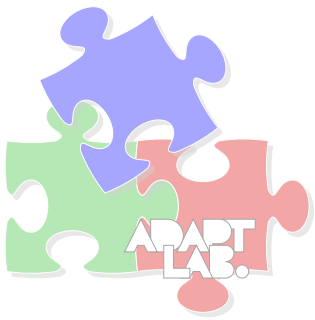
Employing Metaprogramming: A Macro-Driven Strategy to Overcome Compiler-Imposed Limitations on Specialization

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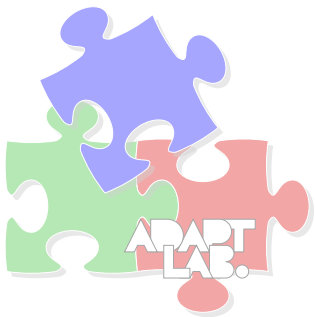


Specialization
State of the Art and Limitations

Specialization aims to combine generic programming with type-specific optimizations.

Mechanisms vary from mature and widely used systems to experimental features, with varying degrees of success.

In many modern ecosystems, specialization is often unstable, forcing a tradeoff between performance and compiler stability.



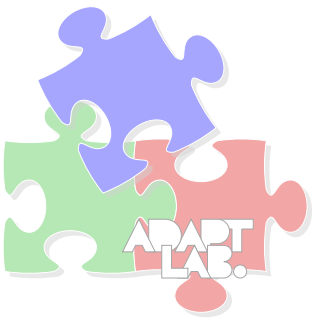


Problem Statement
What's happening in Rust?

In 2015, an RFC introduced specialization to Rust as an unstable feature confined to the Nightly Channel.

However, due to unresolved design issues and concerns about soundness of the type system, specialization is still unstable.

Without stable specialization, developers often resort to code duplication or complex workarounds.

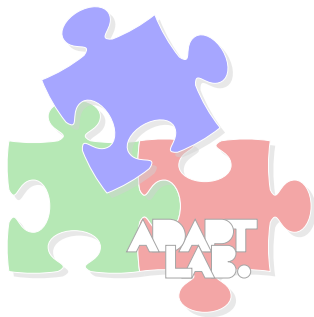




Problem Statement
Motivational Example

Consider implementing a trait for different types, with a specialized implementation for a specific type.

```
1  trait MyTrait { fn my_method(&self); }
2  impl<T> MyTrait for T {
3      fn my_method(&self) {
4          println!("Generic implementation");
5      }
6  }
7  impl MyTrait for i32 {
8      fn my_method(&self) {
9          println!("Specialized for i32");
10     }
11 }
12 fn main() {
13     10.my_method();                // should print "Specialized for i32"
14     10.5.my_method();              // should print "Generic implementation"
15 }
```



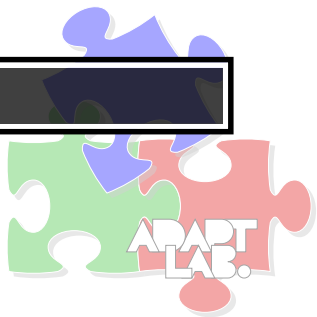


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3      fn my_method(&self) {
4          println!("Generic implementation");
5      }
6  }
7  impl MyTrait for i32 {
8      fn my_method(&self) {
9          println!("Specialized for i32");
10     }
11 }
12 fn main() {
13     10.my_method();                // should print "Specialized for i32"
14     10.5.my_method();              // should print "Generic implementation"
15 }
```

```
error[E0119]: conflicting implementations of trait MyTrait for type i32
```



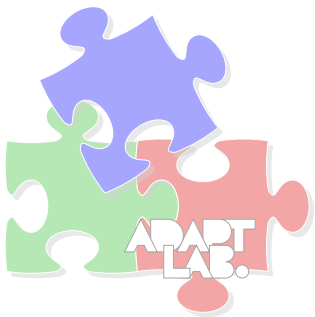


Proposal
Macro-Based Specialization

Employ the powerful metaprogramming capabilities of Rust to build a macro library that simulates native specialization, with:

- Native-like look and feel
- Compile-time resolution
- Stable and sound
- Monomorphization
- Powerful API

To do this, we defined two macros: `when` and `spec`.

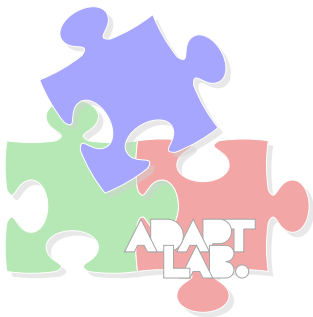




By Example: Implementing the Proposal
Motivational Example

Let's revisit the motivational example we presented earlier.

```
1  trait MyTrait { fn my_method(&self); }
2  impl<T> MyTrait for T {
3      fn my_method(&self) {
4          println!("Generic implementation");
5      }
6  }
7  impl MyTrait for i32 {
8      fn my_method(&self) {
9          println!("Specialized for i32");
10     }
11 }
12 fn main() {
13     10.my_method();                // should print "Specialized for i32"
14     10.5.my_method();              // should print "Generic implementation"
15 }
```

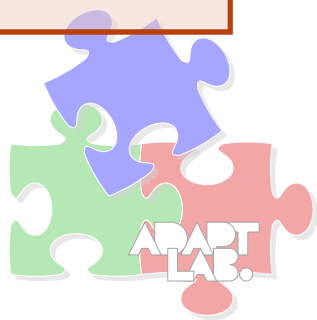




By Example: Implementing the Proposal
Motivational Example

Let's revisit the motivational example we presented earlier.

```
1  trait MyTrait { fn my_method(&self); }
2  impl<T> MyTrait for T {
3      fn my_method(&self) {
4          println!("Generic implementation");
5      }
6  }
7  #[when(T = i32)]
8  impl<T> MyTrait for T {
9      fn my_method(&self) {
10         println!("Specialized for i32");
11     }
12 }
13 fn main() {
14     spec! { 10.my_method(); i32 };        // prints "Specialized for i32"
15     spec! { 10.5.my_method(); f64 };      // prints "Generic implementation"
16 }
```

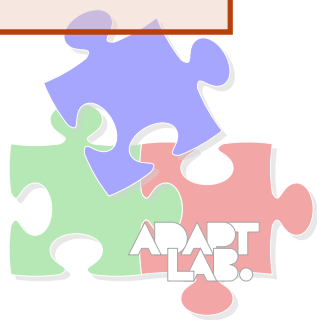




By Example: Implementing the Proposal
Motivational Example

Let's revisit the motivational example we presented earlier.

```
1  trait MyTrait { fn my_method(&self); }
2  impl<T> MyTrait for T {
3      fn my_method(&self) {
4          println!("Generic implementation");
5      }
6  }
7  trait MyTrait_i32 { fn my_method(&self); }
8  impl MyTrait_i32 for i32 {
9      fn my_method(&self) {
10         println!("Specialized for i32");
11     }
12 }
13 fn main() {
14     <i32 as MyTrait_i32>::my_method(&10); // prints "Specialized for i32"
15     <f64 as MyTrait>::my_method(&10.5);   // prints "Generic implementation"
16 }
```





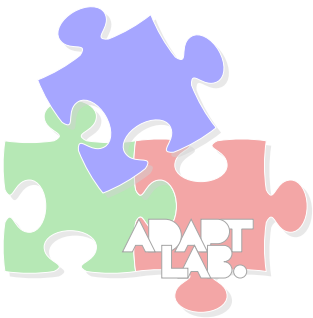
By Example: Implementing the Proposal
Supported Predicates and Forms

The following predicates are supported:

- `T = Type`
- `all(p1, p2, ...)`
- `not(p)`
- `T: Tr1 + Tr2`
- `any(p1, p2, ...)`

We also support the following implementations forms:

- `impl<T> Trait for T`
- `impl<T> Trait for Type<T>`
- `impl<T> Trait<T> for Type`





By Example: Implementing the Proposal
Lifetimes

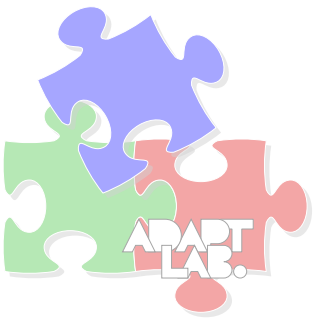
Lifetimes are one of the main reasons native specialization is still unstable.

Lifetimes bounds can be explicit or implicit.

In our belief, this is an architectural limitation of the compiler that can be overcome with proper re-design.

Our proposal treats lifetimes as first-class citizens:

- `T = &'a i32`
- `T: 'a`

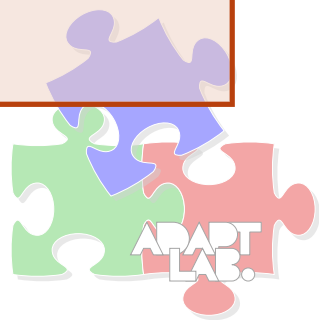




By Example: Implementing the Proposal
Higher-Ranked Polymorphism via HR TBs

Our macro system also supports Higher-Ranked Polymorphism.

```
1  struct S;
2  trait Trait<T, U, V> {
3      fn f(&self, t: T, u: U) -> V;
4  }
5  #[when(all(
6      T = &str, T: 'b,
7      U = for<'a> fn(T, &'a i32) -> V
8  ))]
9  impl<'b, T, U, V> Trait<T, U, V> for S {
10     fn f(&self, t: T, u: U) -> V {
11         u(t, &7)
12     }
13 }
14 fn main() {
15     let s = S;
16     let lambda = |a: &str, b: &i32| a.len() as i32 + *b;
17     let res: i32 = spec! { s.f("abc", lambda); S;
18       [&str, for<'a> fn(&str, &'a i32) -> i32] }; // returns 10
19 }
```

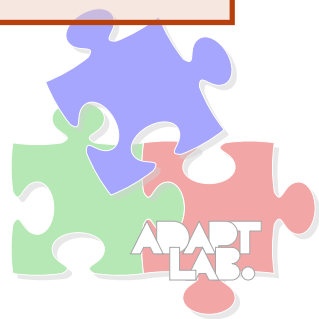




By Example: Implementing the Proposal
Higher-Ranked Polymorphism via HR TBs

Our macro system also supports Higher-Ranked Polymorphism.

```
1  struct S;
2  trait Trait<T, U, V> {
3      fn f(&self, t: T, u: U) -> V;
4  }
5  trait Trait_fn<'b, V> {
6      fn f(&self, t: &'b str, u: for<'a> fn(&'b str, &'a i32) -> V) -> V;
7  }
8  impl<'b, V> Trait_fn<'b, V> for S {
9      fn f(&self, t: &'b str, u: for<'a> fn(&'b str, &'a i32) -> V) -> V {
10         u(t, &7)
11     }
12 }
13 fn main() {
14     let s = S;
15     let lambda = |a: &str, b: &i32| a.len() as i32 + *b;
16     let res: i32 = <S as Trait_fn<_>>::f(&s, "abc", lambda); // returns 10
17 }
```





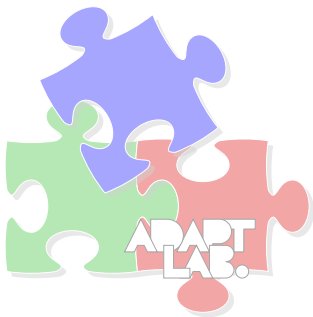
Implementation Details
Handling Conditional Implementations

Conditions defined in the `when` macro are canonized into DNF, flattening when possible, e.g.:

`all(all(T = Vec<U>), any(U = i32, U = f64))` → `any(T = Vec<i32>, T = Vec<f64>)`

A partial order defined over the predicates is used to determine the specificity of each implementation (defined as a relaxation of the lattice-based approach), e.g.:

`T = &i32 > T: Clone || T: Copy`





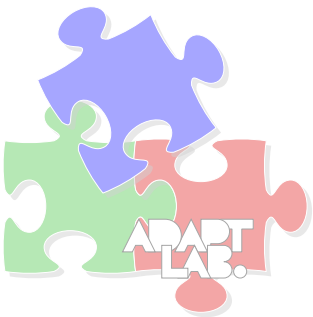
Implementation Details
Handling Conditional Implementations

Due to the lack of reflection, some annotations are required to guide the monomorphization process, e.g.:

```
spec!{ x.f(y); TyX; [TyY]; TyX: Tr1 + Tr2 }
```

We use Build Scripts to collect all the traits defined in the crate and handle the order of their implementations, e.g.:

```
mono_spec::handle_order();
```





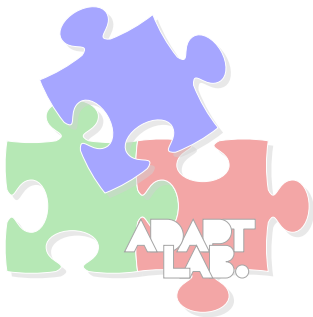
Evaluation
Approach

Once identified some patterns, we instrumented the compiler to analyze at HIR level any function definition.

For each one, we created a custom AST and using novel heuristics, we grouped functions that could be potentially mutually specializable.

We then calculated the tree edit distance using the ZSS algorithm and from that, we derived a similarity score:

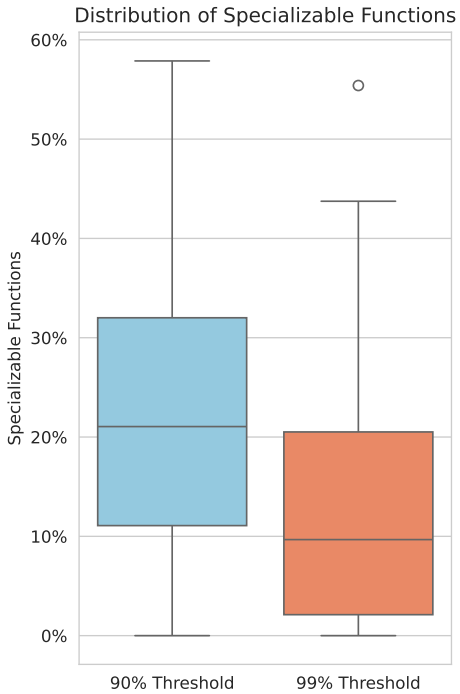
$$1 - \frac{\text{dist}}{\text{max size}}$$

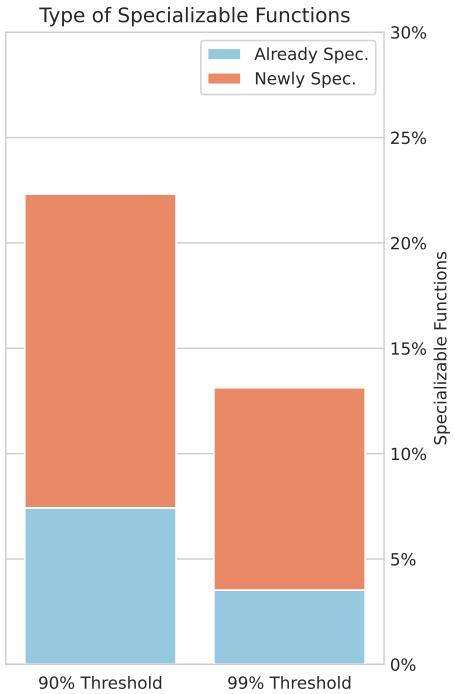


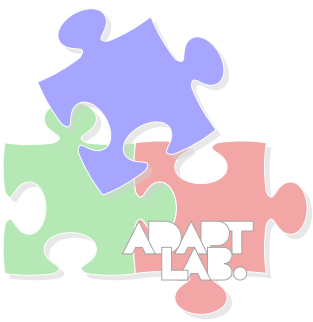


Evaluation
Results

The analysis was run twice on 65 open-source Rust codebases: with a similarity threshold of 90% and 99% respectively.

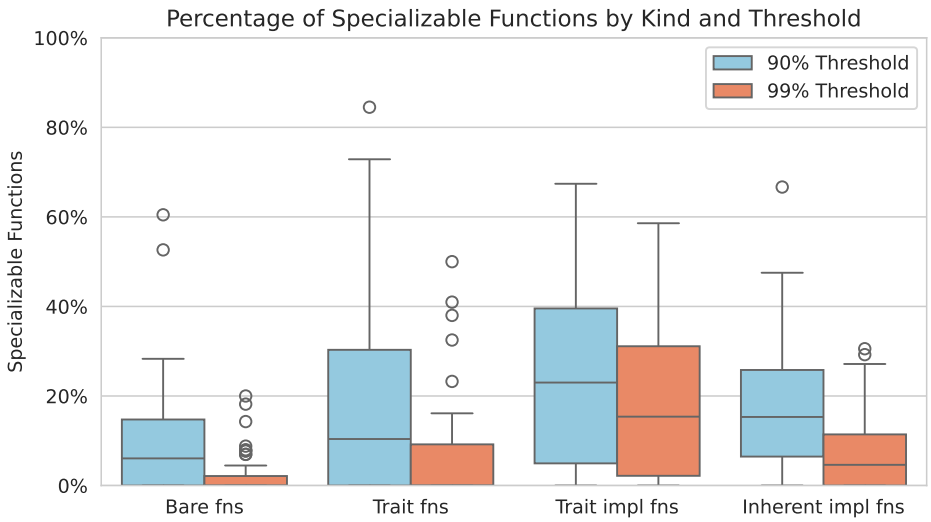


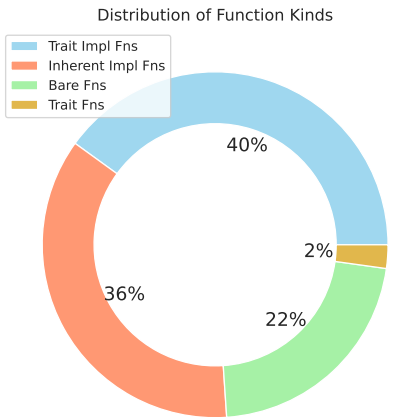




Evaluation
Results

We then analyzed the kinds of functions that were found to be specializable.







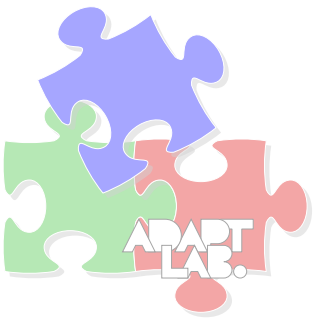
Conclusions
Wrapping up

Our macros bring stable specialization to Rust, enabling more efficient code and improving readability and maintainability.

We are also the only approach to support specialization based on trait bounds, lifetime bounds and negative bounds.

As an extension, we also support overloading, a feature not natively available in Rust.

All of this works without compiler modifications, preserving soundness and ensuring easy adoption by the Rust community.





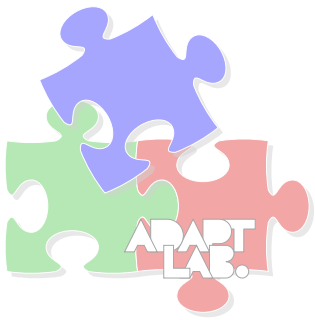
Conclusions
Final notes

Interesting results:

- We've written an article (Meta-Monomorphizing Specializations), submitted to ECOOP.

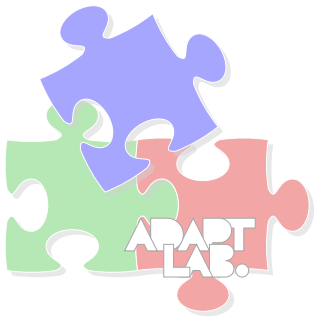
Interesting twist:

- The approach can also be applied to other languages or even directly into the compiler.





Thanks for your attention!





Pattern example
Inherent impl


```
1  struct S;
2  impl S {
3      fn my_method_default<T>(&self, x: T) {
4          println!("Generic implementation");
5      }
6      fn my_method_i32(&self, x: i32) {
7          println!("Specialized for i32");
8      }
9  }
10 fn main() {
11     let s = S;
12     s.my_method_i32(10);         // prints "Specialized for i32"
13     s.my_method_default(10.5);   // prints "Generic implementation"
14 }
```

